RIZAL TECHNOLOGICAL UNIVERSITY
Boni Ave, Mandaluyong, 1550 Metro Manila
College of Engineering

COMPUTER ENGINEERING LAWS & PROFESSIONAL ETHICS
SCOPE OF COMPUTER ENGINEERING DOCUMENTATION

GROUP 2
CEIT-03-602P

Competente, Louis Miguel G.
De Guzman, Jeanette Grace A.
Maylas, Jyanne JM L.

Schedule:
Monday - 4:00 pm to 6:00 pm
Second Semester, 2025-2026

Engr. Julius U. Cabauatan

TABLE OF CONTENTS

1. **Embedded and Real-time Systems**
    a. Definition
    b. Key Characteristics
    c. Types of Real-time Systems
    d. Major Components of Real-time Systems
    e. Examples and Applications
    f. Advantages
    g. Challenges
2. **Software Engineering and System Programming**
    a. Core Areas/Key Concepts
    b. Role in Computer Engineering
    c. Common Applications
3. **Computer Networks and Data Communication**
    a. Core Areas/Concepts
    b. Role in Computer Engineering
    c. Applications
4. **Artificial Intelligence and Machine Learning**
    a. Core Areas/Concepts
    b. Role in Computer Engineering
    c. Common Applications
5. **Cybersecurity and Data Protection**
    a. Core Areas
    b. Role in Computer Engineering
    c. Application
6. **Internet of Things (IoT) and Cloud Computing**
    a. Core Areas
    b. Role in Computer Engineering
    c. Applications
7. **References**

1. Embedded and Real-time Systems

Embedded Systems
An embedded system is a specialized computer system that is intended to perform specific functions within a larger mechanical or electrical system. Unlike general-purpose computers (such as laptops), embedded systems are designed for specific tasks, often with limited hardware resources and little user interaction.
They are "embedded" because they are integrated into a larger mechanical or electronic system. Embedded systems power a wide range of products, machines, and intelligent operations, including machine learning and artificial intelligence applications.

Real-Time Systems
A real-time system is a computing system that must produce accurate results within a specific time frame. It is any information processing system that performs application functions within predictable and precise time limitations, usually on the scale of microseconds. In these systems, timeliness is equally important as accuracy. Missing a deadline can result in system failure or serious consequences.

Key Characteristics of Embedded Systems

Special-Purpose Computing
Embedded systems are designed for one dedicated function rather than multiple general tasks, therefore the hardware and software are optimized for this specific purpose. These are often integrated directly into devices or machines, and to keep them compact, they typically use microcontrollers or system-on-chip (SoC) processors. An example would be a washing machine, as it only manages wash cycles, water levels, and motor control.

Low Power Consumption
As embedded systems are battery-powered and portable devices, they are designed to operate using minimal energy. These systems employ power-saving modes such as sleep or idle states. An efficient hardware and lightweight software package can reduce energy consumption even further. Examples include smartwatches and medical implants that must operate for long periods on small batteries

Time-Critical Processing
Embedded systems must respond to inputs within guaranteed time limits. It uses real-time operating systems (RTOS) for precise task scheduling, which ensures predictable and deterministic system behavior. An example is an airbag controller that must deploy within milliseconds after detecting a collision.

Types of Real-Time Systems

Hard Real-Time Systems

This type of system never misses its deadline. Missing the deadline could have disastrous implications. The usefulness of outputs generated by a hard real-time system drops dramatically and may turn negative as delay increases. Examples include air bags, pacemakers, and aircraft flight control systems.

Firm Real-Time Systems

These are systems that fall somewhere between hard and soft real-time systems. Missing a deadline with firm real-time systems is acceptable, but the output's usefulness diminishes over time. An example is industrial robotic assembly lines, where a delay in schedule can degrade the work quality.

Soft Real-Time Systems

This type of system may occasionally miss its deadline, but with an acceptable low probability. Missing the deadline has no negative effects. Examples include video streaming and online gaming systems. A soft real-time system missing its deadline can manifest itself as latency or lag.

Major Components of Embedded Systems

Processor

This is the embedded system's core computational unit. It executes programmed instructions and can be a low-cost microcontroller, a specialized Digital Signal Processor (DSP) for data streams, or a complex System-on-Chip (SoC) that combines multiple components. The choice affects the system's power and capability.

Memory

Memory holds all critical system data. Non-volatile ROM or Flash memory stores the firmware and permanent program code. Volatile RAM provides a quick, temporary workspace for data and variables during real-time operations. This division ensures consistency and speed.

Input Sensors

Input Sensors are hardware components that detect physical conditions in the environment, such as temperature, motion, or light. They convert these real-world analog signals into digital electrical data. This data is then read and analyzed by the system's processor. Their primary function is to provide the critical information needed for the system to make automated decisions.

Output Actuators

Actuators are devices that perform physical actions in response to processor commands. Examples include motors, valves, lights, and displays. They use electrical signals from

the system to generate kinetic movement, light, sound, or other tangible outputs that affect the environment.

Communication Interfaces

These are the hardware protocols that enable the embedded system to exchange data with other devices. Common standards include Universal Asynchronous Receiver-Transmitter (UART) for simple serial links, Serial Peripheral Interface (SPI) and Inter-Integrated Circuit (I^2C) for onboard chip communication, and Controller Area Network (CAN) for robust automotive or industrial networks, facilitating system connectivity.

Power Supply Unit

This critical subsystem converts an external power source (such as AC mains or a battery) into the stable, regulated DC voltages needed by all other electronic components. It ensures consistent operation by supplying clean power and frequently includes surge or fluctuation protection.

Examples and Applications

In **automotive control systems**, an anti-lock braking system (ABS) monitors a vehicle's individual wheels and pulsates the brakes several times per second to prevent them from locking up during harsh braking. Another system for safety are airbags. Airbags are cushions that inflate immediately on impact (and then deflate) to protect passengers in a crash. They act as a soft restraint between the people and the vehicle interior during a collision, potentially reducing or eliminating injuries. Meanwhile, Engine Control Units (ECU) is a computerized component found in automobiles that controls and maintains numerous systems by receiving real-time data from sensors scattered throughout the vehicle. All of these have to respond in milliseconds to ensure passenger safety.

In **medical devices**, a pacemaker is a small battery-powered device that detects when your heartbeat is too slow. It delivers a signal to your heart, causing it to beat at the proper rhythm. Another system is the infusion pump. These can provide precise volumes of fluids, as well as nutrition and pharmaceuticals. Lastly, MRI (magnetic resonance imaging). It uses a large magnet, radio waves, and a computer to produce clear images of the structures within the body. MRIs are used by healthcare providers to evaluate, diagnose, and monitor a wide range of medical disorders. They all have to be accurate and immediate responses to biological signals.

On **industrial robots**, they are designed to handle automated production activities that need precise motion control and rigorous timing. This feature is based on their capacity to seamlessly coordinate sensors and actuators in real time, allowing for accurate and dependable automation in complicated production situations.

For **smart appliances** such as washing machines, refrigerators, and air conditioners, they use embedded controllers to automate functions and increase efficiency. These intelligent gadgets are typically connected to IoT networks, allowing for remote monitoring, control, and better operational coordination.

Advantages
Embedded systems are small and cost-effective because they are very efficient for the purposes for which they were designed. They can operate on low power thanks to this form factor, which makes them dependable and predictable.

Challenges
While their small form factor makes them compact, it also means that they have limited memory and processing power available. Being built for a specific purpose means that the hardware and software have to be co-designed and specific, adding to complexity. For the embedded systems that deal with safety, they are required to have strict timing constraints and reliability requirements.

2. Software Engineering and System Programming

Software engineering is concerned with the systematic design, development, testing, and maintenance of software systems. It assures that software is high-quality, dependable, and maintainable, particularly for big software systems. It also enhances efficiency, quality, and budget management, allowing software to be delivered on time, within budget, and satisfying all specifications.

System programming deals with low-level software that communicates directly with hardware and operating systems. It is the practice of creating low-level software that interacts directly with computer hardware and operating system kernels to manage resources such as memory, CPUs, and input/output devices, all while providing critical services to higher-level applications.

Core Areas/Key Concepts

Software Development Life Cycle
The Software Development Life Cycle (SDLC) is a systematic framework used by software companies to design, build, and test high-quality software. It defines the complete software development lifecycle, from initial concept to final deployment and maintenance.

The purpose of SDLC is to create high-quality software that meets or exceeds customer expectations, is completed within schedule and budget constraints, and is easy to maintain.

Requirements Analysis and System Design

Requirements analysis is a series of processes that helps define users' expectations for the program you are developing or updating. The process includes assessing, documenting, validating, and maintaining system or software requirements. Requirements analysis includes a variety of tasks that assist engineers in understanding stakeholder demands and explaining them in clear and visual terms.

System design refers to the process of planning, structuring, and defining the architecture of a software system. It entails converting user requirements into a detailed blueprint that will guide the implementation phase. The goal is to create a well-organized and efficient structure that serves its intended purpose while taking into account scalability, maintainability, and performance.

Programming Paradigms

Procedural programming is a programming paradigm based on the concept that programs are sequences of instructions to be performed. They place a strong emphasis on breaking down programs into named sets of instructions known as procedures, which are equivalent to functions. A procedure can hold local data that cannot be accessed from outside its scope, as well as access and modify global data variables.

Object-oriented programming is a programming technique distinguished by the recognition of classes of objects that are inextricably tied to the methods (functions) with which they are related. It also discusses the inheritance of attributes and methods. It is a mathematical technique based on "abstract data types" for storing data and the procedures required to process that data.

Functional programming is a software development methodology that employs pure functions to produce maintainable code. In other words, you create programs by applying and combining functions.

Functional programming leverages language support by employing functions as variables, arguments, and return values, resulting in elegant and clean code. It also uses immutable data and avoids notions such as shared states.

Version control and software project management

Version control, or source control, is the technique of recording and managing changes to software code. Version control systems are software tools that assist development teams in tracking changes to source code over time.

Software project management is the process of planning, arranging, and supervising the creation and delivery of software projects. It entails managing resources, tasks, and timeframes to ensure that the project fulfills its objectives within the parameters of scope, time, cost, and quality.

Operating systems and kernel programming

An **operating system** (OS) is a suite of software that manages a computer's hardware and programs by assigning resources such as memory, CPU, input/output devices, and file storage.

Kernel programming is creating, maintaining, or changing the core, privileged component of an operating system (OS) that serves as a link between software programs and hardware.

Device drivers and embedded software
A **device driver** is a sort of software program or application that manages a specific hardware device and allows other hardware devices to interface with the computer's operating system. A device driver connects with computer hardware via a computer subsystem or a computer bus connected to the hardware.

Embedded software is specialized code or firmware that is designed and incorporated into any device other than a regular computer. It regulates and handles highly specific and limited processes while being intimately linked with the hardware it governs. Embedded software controls the device functions like a computer's operating system controls the function of the software applications.

Compiler and interpreter design
Compiler design involves converting high-level source code into machine code or another intermediate representation that can be executed by a computer. This operation is normally performed once, resulting in a standalone executable file.

Interpreter design focuses on directly executing the source code line by line or statement by statement, rather than creating a separate executable file.

Role in Computer Engineering

Computer engineers bridge hardware and software, ensuring that software efficiently utilizes hardware resources. They design system-level programs such as:

Operating systems
An operating system is the core software that controls all computer hardware and software resources. It provides common services to application programs and serves as a bridge between users and computers. Its main functions are memory management, process scheduling, and file system control.

Firmware for embedded devices
Firmware is low-level software that is permanently programmed into a device's read-only memory (ROM). It provides the necessary control program for the specified hardware. It initializes the device and manages its basic functions without requiring an operating system. This allows devices such as routers and appliances to function autonomously.

Real-time system controllers
These are specialized software systems that must provide a response within a specific, predictable time frame. They monitor inputs and control outputs for time-critical

processes such as car braking and robotic arms. Failure to meet timing deadlines can result in system failure or hazardous conditions.

Hardware abstraction layers

This software layer hides the complexity of specific hardware details from the operating system and applications. It defines a standard interface for software to interact with various hardware components. This enables software to be portable across multiple hardware platforms without requiring extensive rewriting.

Common Applications

Embedded system firmware

It is specialized software that has been permanently programmed into a hardware device's read-only memory. It provides low-level control over the device's specific functions, directly managing its sensors, actuators, and other electronic components. This software is highly optimized for a dedicated system's dependability, real-time performance, and resource efficiency.

Mobile and desktop operating systems

These are comprehensive software platforms that control all hardware and software resources on a computer. They provide essential services such as a user interface, memory management, file system access, and application program security protocols. These systems serve as an intermediary between the user's applications and the device's physical components, such as the processor and memory.

Industrial automation software

It is a comprehensive set of tools and applications used to control and automate industrial processes. This software combines a variety of technologies, such as control systems, robotics, and data analysis tools, to streamline operations, increase productivity, and ensure consistent quality.

Real-time control systems

These are critical in operations where timing is paramount, ensuring that tasks are executed promptly and efficiently. These systems are widely used in a variety of industries, including automotive, aerospace, and manufacturing, and provide precise control and monitoring in dynamic environments.

Enterprise software systems

Also known as enterprise application software (EAS), it is computer software designed to satisfy the needs of an organization rather than individual users. It is a critical component of a computer-based information system, covering a wide range of business processes such as management reporting, production, and back-office duties.

3. Computer Networks and Data Communications

Computer Networks
A computer network is a collection of interconnected computing devices, ranging from traditional to cloud-based, that communicate and share resources with one another. Computer networks are the foundation of almost every digital experience, from personal communication and entertainment to cloud-based commercial operations and global infrastructure.
Data Communication
It is the process of transferring data from one location to another, or between two points. It allows electronic and digital data to be transferred between two networks regardless of their geographical location, data content, or format.

Core Areas/Concepts

Network types/Architectures
Local Area Network
A local area network (LAN) is a network that operates within a small geographic area, typically within the same building. LANs almost always use Ethernet, WiFi, or both in order to connect devices within the network.

Wide Area Network
A wide area network (WAN) is a huge computer network that connects multiple computers across long distances. WANs are often used by large businesses to connect their office networks; each office typically has its own local area network, or LAN, and these LANs connect via a WAN.
Metropolitan Area Network
A metropolitan area network (MAN) is a computer network that connects computers in a metropolitan area, which can be a single large city, a group of cities and towns, or any large area with multiple buildings. A MAN is larger than a LAN, but smaller than a WAN.

Network Protocols

TCP/IP
Transmission Control Protocol (TCP) is a communication standard that allows application programs and computing devices to exchange messages over a network. It is intended to send packets across the internet and ensure the successful transmission of data and messages across networks.
Internet Protocol (IP) is the means for transmitting data from one device to another across the internet. Every device has a unique IP address that allows it to communicate and exchange data with other internet-connected devices.

User Datagram Protocol (UDP)
It is a communication protocol used in time-sensitive applications such as gaming, video streaming, and Domain Name System (DNS) lookups. UDP speeds up communication

because it does not spend time establishing a firm connection with the destination before transferring data.

Hypertext Transfer Protocol (HTTP)
It is the foundation of the World Wide Web, and it is used to load webpages via hypertext links. HTTP is an application layer protocol that runs on top of the network protocol stack. A typical HTTP flow involves a client computer sending a request to a server, which then returns a response message.

File Transfer Protocol (FTP)
It is a common network protocol that transfers files from one host to another across a TCP-based network, such as the Internet. FTP operates by establishing two connections between computers attempting to interact with one another. One connection is reserved for commands and responses exchanged between the two clients, while the other channel is used for data transfer.

OSI reference and TCP/IP models
The **Open Systems Interconnection (OSI) Model** is a set of rules that describe how different computer systems communicate via a network. The OSI Model is divided into seven layers, each with its own set of duties and responsibilities. This layered approach makes it easier for different devices and technologies to work together.
It is composed of:

- **Physical Layer** - it transmits individual bits from one node to the next.
- **Data Link Layer** - is responsible for the node-to-node delivery of the message.
- **Network Layer** - works for the transmission of data from one host to the other located in different networks.
- **Transport Layer** - provides services to the application layer and takes services from the network layer.
- **Session Layer** - is responsible for the establishment of connections, management of connections, terminations of sessions between two devices.
- **Presentation Layer** - The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.
- **Application Layer** - This layer also functions as a window for application services to access the network and display received information to the user.

The **TCP/IP model** is a layered networking framework that describes how data is transferred between devices via a network using established protocols to ensure dependable and efficient transmission. This serves as the foundation for modern Internet and networking technologies.

It is composed of:

- **Application Layer** - the top layer of the TCP/IP model, which is closest to the user, is where applications such as web browsers, email clients, and file-sharing tools communicate with the network.

- **Transport Layer** - it ensures that data is sent reliably and efficiently across devices by controlling segmentation, ordering, and retransmission as needed.
- **Internet Layer** - Responsible for addressing, packing, and routing data packets so that they can transit across networks and reach their intended destination. It guarantees that data moves efficiently across networks.
- **Network Access (Link Layer)** - Responsible for physically transferring data via network gear such as cables, switches, and wireless connections. It manages how data is structured for the network medium and ensures that it reaches the next device along the path.

Routing and switching

Routing is the process of determining the best route for data across one or more networks. It is critical to the internet because it ensures that data packets get to their intended destination in an efficient manner.

Switching is the process of transmitting data from one device to another inside the same network or across multiple networks utilizing switches. It primarily occurs at the OSI model's Data Link Layer (Layer 2), when data is sent in the form of frames using MAC addresses.

Wireless communication/technologies (Wi-Fi, Bluetooth, 5G)

Wi-Fi

It is a wireless networking technology that enables computers (laptops and desktops), mobile devices (smart phones and wearables), and other equipment to connect to the Internet. It enables these devices, and many others, to communicate with one another, forming a network.

Bluetooth

Bluetooth is a wireless technology that allows for short-range voice and data communication. It is a Wireless Personal Area Network (WPAN) technology that allows for data transfers over shorter distances. It operates in the unlicensed, business, scientific, and clinical (ISM) bands between 2.4 GHz and 2.485 GHz.

5G

It represents the fifth generation of cellular technology. It is intended to improve the speed, latency, and flexibility of wireless services. 5G technology offers a theoretical top speed of 20 Gbps, whereas 4G's peak speed is only 1 Gbps. 5G also offers lower latency, which can boost the performance of commercial apps and other digital experiences (such online gaming, videoconferencing, and self-driving cars).

Network security and management

Network security is the protection of the underlying networking infrastructure against illegal access, misuse, or theft. It involves creating a secure infrastructure that allows devices, applications, users, and applications to function securely.

Network security consists of numerous levels of defense both at the edge and within the network. Each network security layer enforces regulations and controls. Authorized

users can access network resources, but malevolent actors are prevented from carrying out vulnerabilities and attacks.

Network management refers to the skills, tools, and processes required to monitor the overall performance of a computer network. Effective network administration guarantees that users have efficient and safe access to networking resources (such as network hardware, storage, memory, bandwidth, data, and processing power).

Role in Computer Engineering

Design network hardware interfaces

Computer engineers create the physical and logical interfaces that enable hardware components to connect and interact via a network. This includes developing electrical specifications, data encoding systems, and connector types for components like as network interface cards (NICs), transceivers, and modem chips. Their work assures consistent data transfer and compatibility with various devices and network standards, such as Ethernet or Wi-Fi. This hardware-level design provides the foundation for all networked systems.

Implement Communication protocols

The role involves creating the software and firmware rules that control how data is formatted, delivered, and received via a network. Protocol stacks (such as TCP/IP) are implemented by computer engineers in both operating system software and specialized chip hardware logic. They follow established international standards to enable robust error handling, data flow control, and effective bandwidth utilization. This solution is critical for facilitating easy and dependable communication between various devices on a local or worldwide scale.

Configure Routers, switches, and network controllers

Configuring routers, switches, and network controllers is the process of establishing and managing the core hardware that governs data flow across all types of networks. Computer engineers design these devices to efficiently route traffic, segment networks for performance and security, and enforce policies that prioritize critical data. This work transforms architectural network designs into stable, secure, and operational realities. Finally, proper configuration ensures that all network users and devices have reliable connectivity and optimal performance.

Develop Embedded networking systems

This task focuses on developing specialized hardware and software that incorporates networking capabilities directly into non-computer devices or dedicated systems. Examples include control modules in industrial IoT sensors, networked automotive systems, and smart home appliances. Engineers must account for constraints such as low power consumption, real-time operation, and a small memory footprint while maintaining secure and stable network connectivity. This advancement is critical to the interconnected, smart functionality of modern embedded devices.

Applications

Internet infrastructure

It is made up of physical and logical systems that enable internet communication among devices, networks, and users. Understanding these layers is critical to comprehending how the internet operates.

The physical layer is the tangible component: cables, servers, satellites, and all of the hardware that transports data from point A to point B. The logical layer consists of protocols and standards (such as TCP/IP, DNS, and routing systems) that direct data where to go and how to get there.

Mobile communication systems

It is a wireless network that allows voice and data communication between devices. Its core infrastructure consists of user handsets, interconnected base stations, and a central switch network. This system manages connectivity by routing calls and data sessions as users move around, resulting in seamless mobile service.

Cloud data centers

A cloud-based data center is a virtualized facility that utilizes cloud computing technology to deliver resources and services via the internet. Unlike traditional data centers, which use physical servers in a single location, cloud-based data centers distribute processing power and storage across a network of servers. This decentralization allows users to access computing resources on demand, thereby increasing efficiency and flexibility.

Smart city networks

Smart city networks are interconnected systems using IoT sensors and data analytics to manage urban infrastructure. They enable real-time monitoring and automated control of services like traffic, energy, and public safety. This connectivity improves efficiency, sustainability, and residents' quality of life.

Industrial network automation

Industrial network automation uses specialized communication protocols to connect machinery, sensors, and control systems throughout a factory. This networked infrastructure allows for real-time monitoring and automated control of production processes. The end result is significantly improved operational efficiency, reliability, and coordination throughout the manufacturing process.

4. Artificial Intelligence and Machine Learning

Software Engineering focuses on the systematic design, development, testing, and maintenance of software systems. System Programming deals with low-level software that directly interacts with hardware and operating systems. **Artificial Intelligence** focuses on creating systems that can perform tasks requiring human intelligence, such as reasoning, learning, and decision-making. **Machine Learning**, a subset of AI, enables systems to learn patterns from data and improve performance over time without being explicitly programmed.

Core Areas

Neural networks are computational models inspired by the human brain that learn to map inputs to outputs through layered units called neurons. **Deep learning** refers to neural networks with many layers that can automatically extract hierarchical features from raw data, which makes them especially powerful for tasks like image recognition and speech processing. These models require substantial training data and computational resources but often deliver state-of-the-art performance in complex pattern-recognition problems.

Natural Language Processing (NLP) enables machines to understand, interpret, and generate human language across tasks such as translation, summarization, sentiment analysis, and conversational AI. NLP systems combine linguistic rules, statistical models, and deep learning to process text and speech, handling ambiguities and context in language. Advances in transformer architectures and large pretrained language models have dramatically improved the fluency and capability of modern NLP applications.

Computer vision equips machines with the ability to extract meaningful information from images and video, enabling tasks like object detection, segmentation, and scene understanding. It relies on techniques ranging from classical image processing to convolutional neural networks and deep learning pipelines that learn visual features directly from pixels. Computer vision powers many real-world systems such as automated inspection, facial recognition, and autonomous navigation.

Expert systems are AI programs that encode domain knowledge and inference rules to make decisions or provide recommendations in specialized areas. They typically use a knowledge base and a rule engine to emulate the reasoning of human experts, making them useful in applications like medical diagnosis, troubleshooting, and regulatory compliance. Although less flexible than learning-based systems, expert systems are valuable where rules must be transparent and justifiable.

Pattern recognition is the process of identifying regularities and structures in data, such as shapes in images, recurring sequences in time series, or clusters in large datasets. It underpins many AI and ML tasks by transforming raw inputs into categories or labels that systems can act upon. Techniques include statistical methods, classical machine learning algorithms, and neural approaches that discover complex patterns automatically.

Reinforcement learning (RL) trains agents to make decisions by interacting with an environment and receiving rewards or penalties based on their actions. RL is particularly suited to sequential decision problems such as game playing, robotics control, and adaptive resource management, where the goal is to learn policies that maximize cumulative reward. Modern RL systems often combine deep neural networks with exploration strategies to scale to high-dimensional tasks.

Role in Computer Engineering

Computer engineers:

Design and optimize specialized hardware, such as GPUs, TPUs, and other accelerators, to speed up large matrix operations central to neural networks. These accelerators provide parallel processing, high memory bandwidth, and specialized instructions that dramatically reduce training and inference times. Engineers balance throughput, power consumption, and cost to make AI computation feasible for both data centers and edge devices.

Optimize ML algorithms for embedded systems. Bringing ML to resource-constrained devices requires algorithmic and software optimizations, including model compression, quantization, and pruning. Computer engineers adapt and tune models so they run efficiently on microcontrollers, mobile SoCs, or edge accelerators without sacrificing essential accuracy. These optimizations enable real-time inference and energy-efficient intelligent behavior in embedded products.

Integrate AI techniques into control systems to enable adaptive, predictive, and autonomous operations in domains such as robotics, industrial automation, and autonomous vehicles. By combining sensors, control logic, and learned models, these systems can react to changing environments and optimize behavior over time. Reliability, safety, and real-time guarantees are central engineering concerns when deploying AI in control loops.

Implement AI solutions that automate tasks like data analysis, decision support, and process control across business and industrial settings. This involves designing data pipelines, model deployment frameworks, monitoring systems, and feedback mechanisms to keep models accurate and reliable in production. Engineers ensure automation is scalable, secure, and aligned with operational constraints.

Common Applications

Autonomous vehicles use a combination of perception (computer vision, LiDAR), prediction (ML models), and planning algorithms to navigate safely without human intervention. AI components detect objects, estimate trajectories, and make split-second driving decisions under uncertain conditions. Ensuring safety, redundancy, and regulatory compliance are major engineering challenges in this domain.

AI systems assist clinicians with **medical diagnosis systems** by analyzing medical images, lab results, and patient histories to detect diseases or predict outcomes early. Techniques such as deep learning for imaging and NLP for electronic health records can improve diagnostic accuracy and speed, while also providing decision support. Clinical deployment requires rigorous validation, explainability, and strong data-privacy protections.

Smart assistants (voice or text) use NLP, speech recognition, and dialogue management to understand user intent and perform tasks such as scheduling, information retrieval, and home control. These systems combine cloud services and on-device models to balance responsiveness with privacy and offline capability. Continuous learning and personalization improve user experience over time.

Robotics leverage AI for perception, planning, manipulation, and human-robot interaction in settings from manufacturing to service industries. ML enables robots to adapt to varied tasks and environments by learning policies and perception models from data or simulation. Engineers integrate sensors, actuators, and AI software to achieve robust, autonomous behavior.

Predictive analytics applies ML to historical and real-time data to forecast equipment failures, customer behavior, and market trends. **Industries** use these insights to optimize maintenance schedules, reduce downtime, personalize services, and improve operational efficiency. Successful deployment combines domain knowledge, quality data, and continuous model monitoring.

5. Cybersecurity and data protection

Cybersecurity ensures the confidentiality, integrity, and availability of digital systems and data. It is also the practice of protecting systems, networks, and programs from digital attacks. These cyberattacks are usually aimed at accessing, changing, or destroying sensitive information; extorting money from users through ransomware; or interrupting normal business processes.

Data protection focuses on privacy and lawful handling of personal information. It is the process of safeguarding data and restoring important information in the event that the data is corrupted, compromised or lost due to cyberattacks, shutdowns, intentional harm or human error. It embraces the technologies, practices, processes and workflows that ensure rightful access to data, so the data is available when it's needed.

Core Areas

Encryption is the process of converting readable data into an encoded form so that only authorized users can access it. **Cryptography** uses mathematical algorithms and keys to secure information, ensuring data confidentiality during storage and transmission. These techniques prevent attackers from understanding or altering sensitive information even if it is intercepted.

Secure system design involves building hardware and software systems with security integrated from the beginning. This approach reduces vulnerabilities by considering threats, risks, and attack methods during the design phase rather than adding security later. It helps protect sensitive information, system resources, and user data against cyberattacks.

Authentication verifies the identity of users, devices, or systems attempting to access resources. **Access control** determines what actions authenticated users are allowed to perform within a system. Together, they ensure that only authorized individuals can access specific data and system functions.

Network security protects communication networks and the data transmitted through them from unauthorized access and attacks. It uses multiple layers of defense, such as firewalls, intrusion detection systems, and traffic monitoring, to secure both internal and external network connections. These measures help maintain system availability and data confidentiality.

Malware analysis is the process of examining malicious software to understand its behavior, origin, and impact. It helps security professionals determine how a system was infected and how the malware operates. This information is used to improve defenses and prevent similar attacks in the future.

Ethical hacking involves legally testing systems for vulnerabilities by simulating real-world attacks. Ethical hackers identify security weaknesses before malicious attackers can exploit them. This proactive approach helps organizations strengthen their security defenses and reduce risks.

Digital forensics focuses on collecting and analyzing digital evidence related to cyber incidents. It is used to investigate security breaches, malware attacks, and other cybercrimes. The findings support incident response, legal investigations, and system recovery efforts.

Data privacy compliance ensures that organizations handle personal data according to laws, regulations, and ethical standards. These rules define how data should be collected, stored, and shared to protect individual privacy. In the Philippines, this is governed by the Data Privacy Act of 2012 (Republic Act 10173).

Role in Computer Engineering

Computer engineers:

Design secure hardware and software systems by building security into systems from the earliest stages of development rather than adding protection afterward. Computer engineers design systems with built-in security features to reduce vulnerabilities. By applying secure-by-design principles, they integrate protection mechanisms such as access controls and secure architectures into the system from the earliest stages. This approach improves long-term system reliability and resistance to attacks.

Implement encryption in devices and networks to protect data stored in devices and transmitted over networks. Encryption ensures that sensitive information remains

confidential even if intercepted. Engineers must balance strong security with system performance and resource limitations.

Develop intrusion detection systems monitor system activity to identify suspicious or malicious behavior. These systems analyze network traffic and system logs to detect attacks early. Early detection allows organizations to respond quickly and minimize damage.

Ensuring security in embedded systems and IoT devices involves protecting both hardware and software components. Engineers implement secure boot mechanisms, encrypted communication, and secure firmware updates. These measures help protect connected devices, which are often targeted due to limited resources and constant connectivity.

Applications

Cybersecurity **secure banking systems** like online banking platforms, financial transactions, and customer information. Techniques such as encryption, multi-factor authentication, and fraud detection systems prevent unauthorized access and financial crimes. These measures maintain trust in digital banking services.

Government systems store sensitive citizen data and critical infrastructure information. Cybersecurity ensures **government data protection** and secure access, data integrity, and protection against cyber threats. Strong security measures support safe and reliable digital public services.

Corporations use cybersecurity infrastructure to protect internal networks, devices, and confidential business data. Security tools such as firewalls, endpoint protection, and access management help prevent data breaches and operational disruptions. This ensures business continuity and data protection.

Digital identity systems verify the authenticity of users accessing online services. Methods such as biometrics, digital certificates, and multi-factor authentication reduce identity fraud. Secure identity verification enables safe access to financial, government, and online platforms.

Secure communication platforms protect messages and shared data from interception. Encryption technologies, including end-to-end encryption, ensure privacy and confidentiality. These platforms are essential for personal communication, corporate collaboration, and secure information exchange.

6. Internet of Things (IoT) and cloud computing

IoT connects physical devices to the internet for data collection and control.The IoT refers to a network of interconnected physical objects such as devices, machines, vehicles, or people embedded with sensors, software, and unique identifiers that enable them to collect, exchange, and process data over a network without requiring direct human-to-human or human-to-computer interaction. One component that improves the success of the Internet of Things is **Cloud Computing.**

Cloud computing provides on-demand computing resources and storage over the internet. Cloud computing enables users to perform computing tasks using services provided over the Internet. The use of the Internet of Things in conjunction with cloud technologies has become a kind of catalyst: the Internet of Things and cloud computing are now related to each other. These are true technologies of the future that will bring many benefits.

Core Areas

Internet of Things (IoT):
Sensors and actuators are essential components of IoT systems that enable interaction between the physical and digital worlds. Sensors collect real-world data such as temperature, humidity, motion, pressure, or light and convert it into digital signals that embedded systems can process. Actuators perform physical actions based on processed data or control commands, such as turning motors, opening valves, or switching lights on and off. In IoT and cloud-connected systems, sensors provide continuous data to cloud platforms for analysis, while actuators execute decisions made either locally or through cloud-based control systems.

Microcontrollers and embedded systems form the foundation of IoT technology. They act as the “brains” of IoT devices, controlling sensors, actuators, and communication modules. Embedded systems are designed for specific tasks such as data collection, device control, and local processing, while microcontrollers execute firmware that enables devices to interact with their environment. In IoT and cloud computing, these systems gather real-world data and transmit it to cloud platforms for storage, analysis, and remote monitoring. Their low power consumption, real-time operation, and reliability make them essential for scalable and efficient IoT solutions.

Wireless communication protocols enable IoT devices to transmit data to gateways, cloud platforms, and other devices without physical connections. These protocols define how data is formatted, transmitted, and received reliably over wireless networks. Common protocols such as Wi-Fi, Bluetooth, and cellular networks allow IoT devices to communicate over short or long distances depending on application needs. Efficient wireless communication is essential for real-time monitoring, remote control, and

scalable IoT systems, especially in environments where wired connections are impractical.

Edge computing is a computing approach where data processing is performed close to the source of data generation rather than relying entirely on centralized cloud servers. In IoT systems, edge devices such as gateways or embedded controllers analyze and filter data locally, reducing latency, bandwidth usage, and dependence on constant internet connectivity. This approach enables faster decision-making, improved system responsiveness, and enhanced reliability, particularly for time-critical applications such as industrial automation, healthcare monitoring, and autonomous systems.

Cloud Computing:
Virtualization is a fundamental technology in cloud computing that allows multiple virtual machines to run on a single physical server. By abstracting hardware resources such as CPU, memory, and storage, virtualization improves resource utilization, scalability, and flexibility. This enables cloud providers to efficiently allocate computing resources to multiple users while reducing hardware costs.

Cloud service models define how computing resources are delivered to users. Infrastructure as a Service (IaaS) provides virtualized hardware resources such as servers, storage, and networks. Platform as a Service (PaaS) offers a development environment with tools and frameworks for building and deploying applications without managing underlying infrastructure. Software as a Service (SaaS) delivers fully functional software applications over the internet, allowing users to access services without installation or maintenance.

Distributed computing is a computing approach where tasks and data processing are spread across multiple interconnected systems rather than a single machine. In cloud environments, distributed computing improves performance, reliability, and scalability by allowing workloads to be processed in parallel across different servers. This approach is essential for handling large-scale data processing, high-traffic applications, and real-time services.

Data storage and backup in cloud computing provide secure and scalable solutions for storing large volumes of data. Cloud storage systems allow data to be accessed remotely, shared across applications, and expanded as needed. Backup services ensure data protection by creating copies that can be restored in case of data loss, system failure, or cyberattacks, ensuring data availability and reliability.

Role in Computer Engineering
Computer engineers:

Design IoT hardware by integrating microcontrollers, sensors, actuators, and communication modules into a single functional device. They ensure that the hardware can reliably collect data from the environment, process it efficiently, and communicate with other systems or cloud platforms while meeting constraints such as low power consumption and small size.

Firmware development involves writing low-level software that controls how sensors collect data, how devices communicate, and how power is managed. Computer engineers ensure that sensor networks operate reliably, respond to real-world events in real time, and maintain stable communication between multiple interconnected devices.

Connect embedded devices to cloud platforms using networking protocols and APIs. This allows sensor data to be stored, monitored, and analyzed remotely. Cloud connectivity enables remote control, data analytics, system updates, and scalability for large IoT deployments.

Optimize how data is transmitted and processed to reduce latency, bandwidth usage, and energy consumption. Techniques such as data compression, efficient communication protocols, and edge computing help improve performance while ensuring reliable and fast system responses.

Applications

Smart homes use IoT devices or **appliances** such as smart lights, thermostats, and security systems to automate household functions. These devices collect data through sensors and communicate with cloud platforms, allowing users to monitor and control appliances remotely while improving comfort, safety, and energy efficiency.

Wearable health monitor devices like fitness trackers and smartwatches continuously monitor health parameters such as heart rate, steps, and sleep patterns. The collected data is transmitted to cloud servers for analysis, enabling users and healthcare professionals to track health conditions in real time.

In **industrial** environments, **IoT systems monitor machines,** production lines, and equipment performance. Sensors collect real-time data, which is processed locally or in the cloud to detect faults, schedule maintenance, and improve productivity while reducing downtime.

Smart agriculture systems use IoT sensors to monitor soil moisture, temperature, humidity, and crop health. Cloud-based analytics help farmers make informed decisions

about irrigation, fertilization, and harvesting, leading to improved crop yield and efficient resource usage.

Businesses use cloud platforms to store, process, and analyze large volumes of data generated by IoT systems. These platforms support scalability, data analytics, remote access, and cost-effective infrastructure for enterprise operations and decision-making.

References

Ainaa, M. (2025, February 28). What is a Cloud-Based Data Center? *Data Center Asia*. https://www.datacenter-asia.com/blog/what-is-a-cloud-based-data-center/

Airbags: What are they and how do they work? (n.d.). *RoadSafetyFacts.Eu*. Retrieved February 1, 2026, from https://roadsafetyfacts.eu/airbags-what-are-they-and-how-do-they-work/

Andersen, G. (2024a, January 19). *The Role of Computer Engineers in Creating Safe and Secure Systems*. https://moldstud.com/articles/p-the-role-of-computer-engineers-in-building-safe-and-secure-systems

Andersen, G. (2024b, January 21). *The Role of Embedded Software Engineers in Ensuring Cybersecurity in IoT*. https://moldstud.com/articles/p-the-role-of-embedded-software-engineers-in-ensuring-cybersecurity-in-iot

Atlassian. (n.d.). *What is version control | Atlassian Git Tutorial*. Atlassian. Retrieved February 1, 2026, from https://www.atlassian.com/git/tutorials/what-is-version-control

Authentication and Access Control: Everything You Need to Know When Assessing Authentication and Access Control Skills. (n.d.). Retrieved February 1, 2026, from https://www.alooba.com/skills/concepts/information-security-architecture-169/authentication-and-access-control/

Car Anti-Lock Braking Systems (ABS) Explained | Mazda USA. (n.d.). Retrieved February 1, 2026, from https://www.mazdausa.com/resource-center/what-is-abs

Computer Vision Tutorial. (12:58:17+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-vision/computer-vision/

Definition of Object-oriented Programming—Gartner Information Technology Glossary. (n.d.). Gartner. Retrieved February 1, 2026, from https://www.gartner.com/en/information-technology/glossary/oop-object-oriented-programming

Difference between Sensor and Actuator. (15:26:30+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/electronics-engineering/difference-between-sensor-and-actuator/

Digital Identity: The Ultimate Guide to Modern Authentication. (n.d.). Retrieved February 1, 2026, from https://bisresearch.com/news/digital-identity-the-ultimate-guide-to-modern-authentication-and-cybersecurity

EC-Council. (2024, January 19). What is Ethical Hacking. *Cybersecurity Exchange*. https://www.eccouncil.org/cybersecurity-exchange/ethical-hacking/what-is-ethical-hacking/

Edge Computing. (06:57:56+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/edge-computing/

Embedded software: Definition, examples, types | Endava. (n.d.). Retrieved February 1, 2026, from https://www.endava.com/glossary/embedded-software

Expert Systems in AI. (00:10:58+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/artificial-intelligence/expert-systems/

Internet of Things and careers in IoT. (01:49:02+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/internet-of-things-and-careers-in-iot/

Introduction to Cloud Computing. (18:59:24+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/cloud-computing/cloud-computing/

Introduction to Internet of Things—IOT. (05:45:03+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/introduction-to-internet-of-things-iot-set-1/

Introduction to Software Engineering. (05:05:34+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/software-engineering/software-engineering-introduction-to-software-engineering/

IoT and Cloud Computing. (17:56:12+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/cloud-computing/iot-and-cloud-computing/

Lee, S. (n.d.). *8 Applications Driving Cybersecurity Enhancements in Banking Today*. Retrieved February 1, 2026, from https://www.numberanalytics.com/blog/8-cybersecurity-applications-enhancing-banking-safety

Machine Learning Tutorial—GeeksforGeeks. (n.d.). Retrieved February 1, 2026, from https://www.geeksforgeeks.org/machine-learning/machine-learning/

McCarthy, N. (2025, April 28). Enterprise Software Meaning. *Planet Crust*. https://www.planetcrust.com/enterprise-software-meaning/?utm_campaign=blog

Mitchell, B. (2022, July 13). What Is Functional Programming and Why Use It? *Coding Dojo*. https://www.codingdojo.com/blog/what-is-functional-programming

Natural Language Processing (NLP)—Overview. (20:52:22+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/nlp/natural-language-processing-overview/

Pattern Recognition. (n.d.). GeeksforGeeks. Retrieved February 1, 2026, from https://www.geeksforgeeks.org/quizzes/pattern-recognition/

PNA. (2025, March 4). PNPKI ensures cybersecurity in the digital future. *Philippine Information Agency*. https://pia.gov.ph/press-release/pnpki-ensures-cybersecurity-in-the-digital-future/

Procedural Programming • DeepSource. (n.d.). DeepSource. Retrieved February 1, 2026, from https://deepsource.com/glossary/procedural-programming

Real Time Systems. (12:43:39+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-science-fundamentals/real-time-systems/

Roy, R. (n.d.). *What is software project management? Definition, Planning and Best Practices*. Plane. Retrieved February 1, 2026, from https://plane.so/blog/what-is-software-project-management

Secure System Design: Everything You Need to Know When Assessing Secure System Design Skills. (n.d.). Retrieved February 1, 2026, from https://www.alooba.com/skills/concepts/information-security-architecture-169/secure-system-design/

system. (n.d.). *Systems programming*. Grokipedia. Retrieved February 1, 2026, from https://grokipedia.com/page/Systems_programming

System Analysis—SEBoK. (n.d.). Retrieved February 1, 2026, from https://sebokwiki.org/wiki/System_Analysis

System Design Introduction—LLD & HLD. (00:10:47+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/system-design/getting-started-with-system-design/

TCP/IP Model. (13:33:57+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/tcp-ip-model/

What is a Compiler? | IBM. (2025, July 18). https://www.ibm.com/think/topics/compiler

What is a LAN (local area network)? | Cloudflare. (n.d.). Retrieved February 1, 2026, from https://www.cloudflare.com/learning/network-layer/what-is-a-lan/

What is a metropolitan area network (MAN)? (n.d.). Retrieved February 1, 2026, from https://www.cloudflare.com/learning/network-layer/what-is-a-metropolitan-area-network/

What is a WAN? | Wide area networks vs LAN explained | Cloudflare. (n.d.). Retrieved February 1, 2026, from https://www.cloudflare.com/learning/network-layer/what-is-a-wan/

What is an Electronic Control Unit in Automotive Systems? (2024, May 15). Total Phase Blog. https://www.totalphase.com/blog/2024/05/what-is-electronic-control-unit-automotive-systems/

What is an Operating System? | IBM. (2025, March 5). https://www.ibm.com/think/topics/operating-systems

What is Bluetooth? (06:47:27+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/bluetooth/

What Is Computer Networking? | IBM. (n.d.). Retrieved February 1, 2026, from https://www.ibm.com/think/topics/networking

What is cybersecurity? (n.d.). Cisco. Retrieved February 1, 2026, from https://www.cisco.com/site/us/en/learn/topics/security/what-is-cybersecurity.html

What Is Data Compliance? | IBM. (2023, November 8). https://www.ibm.com/think/topics/data-compliance

What is Data Protection and Why is it Important? A Guide. (n.d.). Search Data Backup. Retrieved February 1, 2026, from https://www.techtarget.com/searchdatabackup/definition/data-protection

What is digital forensics? | IBM. (2024, February 16). https://www.ibm.com/think/topics/digital-forensics

What is File Transfer Protocol (FTP) meaning. (n.d.). Fortinet. Retrieved February 1, 2026, from https://www.fortinet.com/resources/cyberglossary/file-transfer-protocol-ftp-meaning

What is HTTP? | Cloudflare. (n.d.). Retrieved February 1, 2026, from https://www.cloudflare.com/learning/ddos/glossary/hypertext-transfer-protocol-http/

What Is Industrial Automation Software? | Key Features. (2024, September 25). https://www.indmall.in/faq/what-is-industrial-automation-software/

What is Network Management? | IBM. (2022, October 20). https://www.ibm.com/think/topics/network-management

What Is Network Security? (n.d.). Cisco. Retrieved February 1, 2026, from https://www.cisco.com/site/us/en/learn/topics/security/what-is-network-security.html

What is Network Security? (01:45:43+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/network-security/

What Is Network Switching? (n.d.). Cisco. Retrieved February 1, 2026, from https://www.cisco.com/site/us/en/learn/topics/networking/what-is-network-switching.html

What is OSI Model | 7 Layers Explained. (20:59:42+00:00). GeeksforGeeks. https://www.geeksforgeeks.org/computer-networks/open-systems-interconnection-model-osi/

What is Software Engineering? (n.d.). Michigan Technological University. Retrieved February 1, 2026, from https://www.mtu.edu/cs/undergraduate/software/what/

What is TCP/IP in Networking? (n.d.). Fortinet. Retrieved February 1, 2026, from https://www.fortinet.com/resources/cyberglossary/tcp-ip

What Is User Datagram Protocol (UDP)? (n.d.). Fortinet. Retrieved February 1, 2026, from https://www.fortinet.com/resources/cyberglossary/user-datagram-protocol-udp

What Is Wi-Fi? - Definition and Types. (n.d.). Cisco. Retrieved February 1, 2026, from https://www.cisco.com/site/us/en/learn/topics/networking/what-is-wi-fi.html